

# Evaluate limits using L'hospital's rule

## Basic skills

Q1. a)  $2x - 5 \times 4x^3 = 2x - 20x^3$

b)  $(x+2)^{\frac{1}{3}} + 2 = \frac{1}{3}(x+2)^{-\frac{1}{3}}$  chain

c)  $\sin x \cos x = 2 \sin x \cos x$  Product

d)  $\ln^2 x = \frac{2}{x} \ln x$

Q2.  $f(x) = x^2 \sin x$   $g(x) = \ln x$   
↓ Product

a)  $f'(x) = 2x \sin x + x^2 \cos x$   $g'(x) = \frac{1}{x}$

b)  $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} x^2 (2 \sin x + x \cos x)$   
L'hospital's

c)  $\lim_{x \rightarrow 0} x^2 (2 \sin x + x \cos x) = 0^2 \times (0 + 0) = 0$

so  $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 0$

Q3.  $\lim_{x \rightarrow \infty} \frac{3+5x}{6x-5} = \lim_{x \rightarrow \infty} \frac{5}{6} = \frac{5}{6}$

L'Hopital's

## Competency skills

Q4. Not necessarily

e.g.  $\lim_{x \rightarrow \infty} \frac{x^2+2x}{3x^2+5} = \lim_{x \rightarrow \infty} \frac{2x+2}{6x}$

$= \lim_{x \rightarrow \infty} \frac{2}{6} = \frac{1}{3}$

Two applications needed

Q5.  $L = \lim_{x \rightarrow 1} \frac{4x^2-7x+3}{\cos(\frac{\pi}{2}x)}$

a) have  $\frac{0}{0}$  problem

b)  $L = \lim_{x \rightarrow 1} \frac{8x-7}{-\frac{\pi}{2} \sin(\frac{\pi}{2}x)} = \lim_{x \rightarrow 1} \frac{8}{-\frac{\pi^2}{4} \cos(\frac{\pi}{2}x)}$

$= \frac{8}{\infty} = 0$

$$\text{Q6. a) } \lim_{x \rightarrow 0} \frac{e^{2x} - x}{e^x - 1} = \lim_{x \rightarrow 0} \frac{2e^{2x} - 1}{e^x}$$

$$= \lim_{x \rightarrow 0} (2e^x - e^{-x}) = 2 - 1 = 1$$

$$\text{b) } \lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2} = \lim_{x \rightarrow \infty} \frac{3e^{3x}}{2x} = \lim_{x \rightarrow \infty} \frac{9e^{3x}}{2} = \infty$$

$$\text{c) } \lim_{x \rightarrow -\infty} \frac{e^{-x}}{x^3 + 2x} = \lim_{x \rightarrow -\infty} \frac{(-1)^3 e^{-x}}{3 \times 2 \times 1} = 0$$

→ apply 3 times

$$\text{d) } \lim_{x \rightarrow \infty} \frac{x^3 - x + 5}{x^2 - 6x + 1} = \lim_{x \rightarrow \infty} \frac{3 \times 2x}{2 \times 1} = \infty$$

→ apply twice

$$\text{e) } \lim_{x \rightarrow 0} \frac{(6 - 5x^2) \sin x}{x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{-10x \sin x + (6 - 5x^2) \cos x}{\cos x - x \sin x} = \frac{6}{1} = 6$$

$$f) \lim_{x \rightarrow \infty} \frac{x^2}{e^{x^2}} = \lim_{x \rightarrow \infty} \frac{\cancel{2x}}{\cancel{2x}e^{x^2}} \\ = \lim_{x \rightarrow \infty} \frac{1}{e^{x^2}} = 0$$

$$Q7. \lim_{x \rightarrow \infty} \frac{x^2 + \sin x}{5x^2 - 3} \text{ currently } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{2x + \cos x}{10x} \xrightarrow{\text{still } \frac{\infty}{\infty}} = \lim_{x \rightarrow \infty} \frac{2 - \sin x}{10}$$

no limit of  $\sin x$  as  $x \rightarrow \infty$  due to oscillation

## Advanced skills

$$Q8. \text{ case 1: } f(x) = a_1 x^n + \dots + a_{n+1} \\ g(x) = b_1 x^n + \dots + b_{n+1}$$

L'Hopital's:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{n! a_n}{n! b_n} = \frac{a_n}{b_n}$$

$\rightarrow$  n l'hopital applications

Division!

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{a_1 + \frac{a_2}{x} + \dots + \frac{a_{n+1}}{x^n}}{b_1 + \frac{b_2}{x} + \dots + \frac{b_{n+1}}{x^n}} = \frac{a_1}{b_1}$$

$\div x^n$

Case 2:

$$f(x) = a_1 x^{n+k} + \dots + a_{k+1} x^n + \dots + a_{n(k+1)}$$

$$g(x) = b_{k+1} x^n + \dots$$

L'hôpital's!

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{(n+k) \dots (k+1) a_1 x^k + \dots}{n! b_{k+1}}$$

$n$  l'hôpital applications

$$= \infty$$

Division!

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{a_1 x^k + \dots + a_{k+1} + \frac{a_{k+2}}{x} + \dots}{b_{k+1} + \frac{b_{k+2}}{x} + \dots}$$

$\div x^n$

$$= \infty$$

Case 3:

$$f(x) = a_{k+1} x^n + \dots$$

$$g(x) = b_1 x^{n+k} + \dots + b_{k+1} x^n + \dots$$

L'Hopital's

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{0}{b_1(n+k)!} = 0$$

$n+k$  applications

Division:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\frac{a_{k+1}}{x^k} + \dots}{b_1 + \frac{b_2}{x} + \dots} = \frac{0}{b_1} = 0$$

$\div x^{n+k}$

Q9. a)  $\lim_{x \rightarrow 0} \frac{\ln(\cos 2x)}{e^x - 1 - x}$

$$= \lim_{x \rightarrow 0} \frac{\frac{-2 \sin 2x}{\cos 2x}}{e^x - 1} = \lim_{x \rightarrow 0} \frac{-4 \sec^2 2x}{e^x}$$

$$= \frac{-4}{1} = -4$$

b)  $\lim_{x \rightarrow 0} \frac{\ln(1 + (-\frac{(2x)^2}{2} + \dots))}{(1 + x + \frac{x^2}{2} + \dots) - 1 - x}$

$$= \lim_{x \rightarrow 0} \frac{(-2x^2 + \dots) + (-2x^2 + \dots)^2 + \dots}{\frac{x^2}{2} + \dots}$$

$$= \lim_{x \rightarrow 0} \frac{-2x^2 + \dots}{\frac{x^2}{2} + \dots} = \lim_{x \rightarrow 0} \frac{-2 + \dots}{\frac{1}{2} + \dots}$$

$$= \frac{-2}{\frac{1}{2}} = -4$$

c) Both were about as quick as each other, but series is more useful as no need for repeated differentiation for a lot of limits